

Geospatial Feature Extraction Platform

Internship Assignment Report

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1 Introduction

This project presents a full-stack geospatial analysis platform developed as part of an internship assignment for TerraAqua UAV. The goal of the system is to process satellite imagery and automatically detect and classify key geographical features such as water bodies, vegetation, and roads.

The system converts raw satellite images into structured geospatial data in the form of GeoJSON and provides both map-based and image-based visualization.

2 System Architecture and Workflow

The system follows a multi-tier architecture:

- **Frontend:** Built using React.js for user interaction and visualization.
- **Java Backend:** Acts as an API gateway using Spring Boot.
- **Python Backend:** Implements the core image processing pipeline using FastAPI.

Workflow:

1. User uploads a GeoTIFF satellite image.
2. The request is sent from React frontend to Java backend.
3. Java backend forwards the request to the FastAPI service.
4. Python backend processes the image and detects features.
5. GeoJSON and processed image are returned to frontend.
6. Results are displayed as map view and image overlay.

3 Methodology

The feature detection is based on classical image processing techniques:

3.1 Preprocessing

- Image normalization using OpenCV.
- Resizing for faster computation.

3.2 Band Analysis

- RGB bands are analyzed dynamically.
- Bands are sorted based on intensity.

3.3 Feature Detection

Water Detection:

- Identified using low intensity values.
- Thresholding applied on dark bands.

Vegetation Detection:

- Detected using high reflectance in mid bands.
- Percentile-based thresholding.

Road Detection:

- Edge detection using Canny algorithm.
- Morphological operations for refinement.

3.4 Post Processing

- Noise removal using morphological operations.
- Overlap removal between features.

3.5 GeoJSON Generation

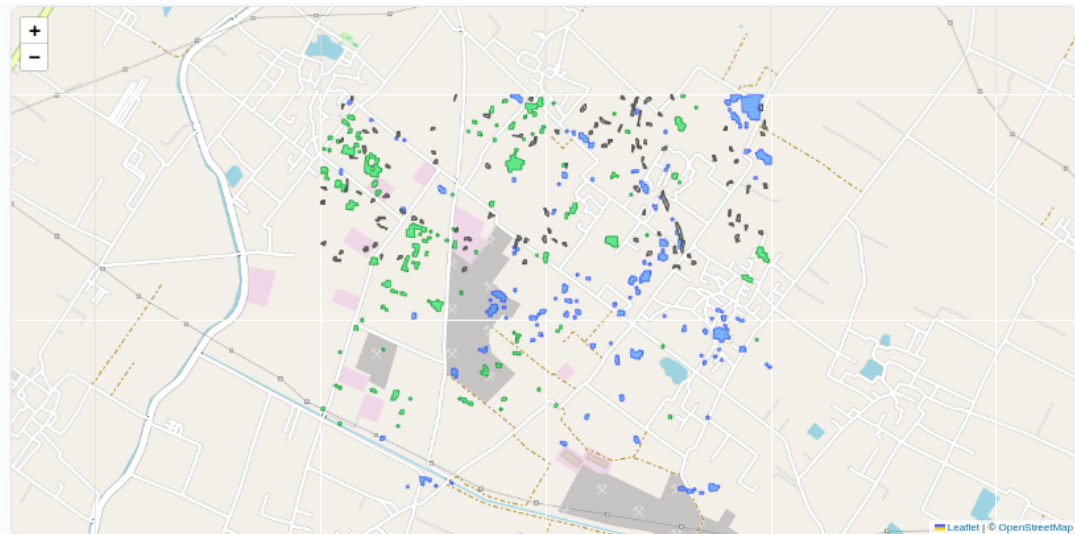
- Masks converted to polygons using Rasterio.
- CRS transformation applied using PyProj.
- Each feature contains:
 - feature_type
 - status
 - area_sq_m

4 Results

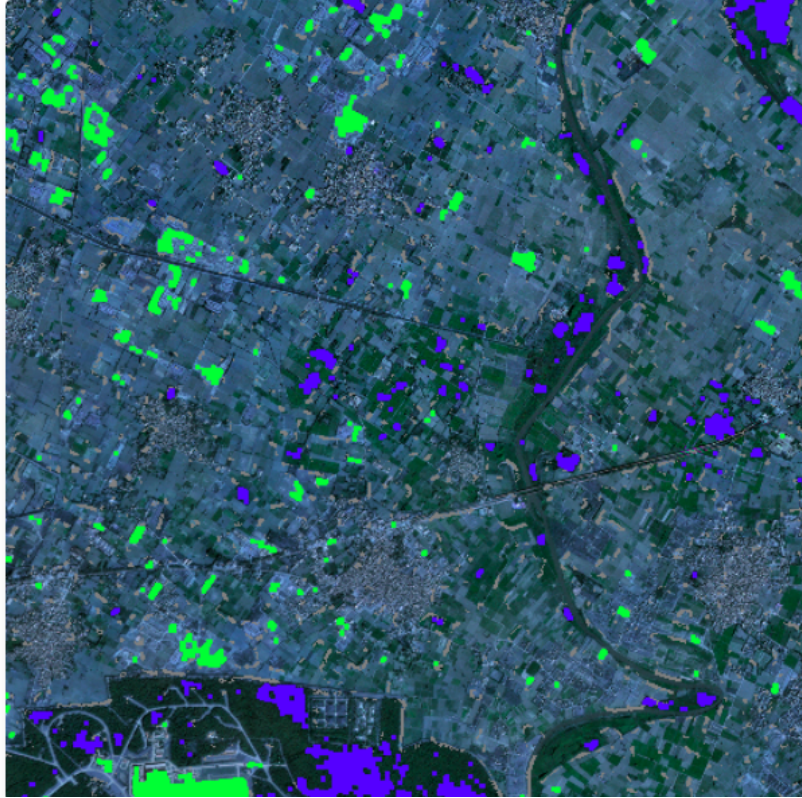
The system successfully detects and visualizes:

- Water bodies (highlighted in blue)
- Vegetation (highlighted in green)
- Roads (highlighted in gray)

4.1 Map Output



4.2 Processed Image Output



5 Observations

- The system performs well on medium-resolution satellite images.
- Water and vegetation detection are relatively accurate.
- Road detection is sensitive to noise and requires fine-tuning.
- Rule-based detection is fast but less robust compared to deep learning methods.

6 Limitations

- No deep learning-based segmentation.
- Accuracy depends on lighting and image quality.
- Limited scalability for very large datasets.

7 Future Improvements

- Integration of deep learning models (CNN/UNet).
- Support for multi-spectral satellite data.

- Cloud deployment for scalability.
- Real-time geospatial analysis.

8 Conclusion

This project demonstrates a complete geospatial processing pipeline from raw satellite imagery to structured GeoJSON outputs. The integration of React, Spring Boot, and FastAPI provides a scalable and modular system suitable for real-world applications.